

Databases Assignment 2 Data Modeling

Database normalization

shelter_id	shelter_name	shelter_address	shelter_capacity	animal_id	animal_name	animal_species	animal_age	animal_sex	animal_breed	intake_id	intake_date	staff_id	staff_name	staff_role
1	Happy Tails	MainStreet 1	60/100	1	Max	dog	2 years	male	border collie	3	03-03-2024	1	Kiki	Vet
1	Happy Tails	MainStreet 1	60/100	2	Gerard	cat	1 year	male	ragdoll	4	27-05-2025	2	Lara	Vet
2	City Paws	Citystreet 3	85/200	3	Luna	dog	7 years	female	golden retriever	5	05-03-2022	3	Zara	Vet
1	Happy Tails	MainStreet 1	60/100	4	Harold	bunny	9 years	male	californian	6	06-03-2026	4	Chantal	Receptionist
1	Happy Tails	MainStreet 1	60/100	5	Simba	bunny	2 years	female	netherland dwarf	7	06-10-2023	5	Ariana	Assistant Vet
1	Happy Tails	MainStreet 1	60/100	6	Kitty	cat	5 years	female	persian	8	04-02-2026	6	Harry	Caretaker
2	City Paws	Citystreet 3	43/200	7	Stella	dog	10 years	female	labradore	9	11-01-2024	7	Benito	Animal Groomer

Table 1 violates 1NF since the cells in capacity store multiple values, the current capacity and the maximum capacity of the shelter. By splitting this into two columns we can fix this.

shelter_id	shelter_name	shelter_address	shelter_capacity	shelter_capacity_max	animal_id	animal_name	animal_speci	animal_age	animal_sex	animal_bre	intake_id	intake_date	staff_id	staff_name	staff_role
1	Happy Tails	MainStreet 1	60	100	1	Max	dog	2 years	male	border collie	3	03-03-2024	1	Kiki	Vet
1	Happy Tails	MainStreet 1	60	100	2	Gerard	cat	1 year	male	ragdoll	4	27-05-2025	2	Lara	Vet
2	City Paws	Citystreet 3	85	200	3	Luna	dog	7 years	female	golden retriever	5	05-03-2022	3	Zara	Vet
1	Happy Tails	MainStreet 1	60	100	4	Harold	bunny	9 years	male	californian	6	06-03-2026	4	Chantal	Receptionist
1	Happy Tails	MainStreet 1	60	100	5	Simba	bunny	2 years	female	netherland dwa	7	06-10-2023	5	Ariana	Assistant Vet
1	Happy Tails	MainStreet 1	60	100	6	Kitty	cat	5 years	female	persian	8	04-02-2026	6	Harry	Caretaker
2	City Paws	Citystreet 3	43	200	7	Stella	dog	10 years	female	labradore	9	11-01-2024	7	Benito	Animal Groomer

Table 2 now satisfies 1NF. However, it does violate 2NF. The primary key is a composite key consisting of shelter_id, animal_id, intake_id, and staff_id. Some non-key attributes depend only partially on this composite key, for example animal_name, animal_species only depend on animal_id. So, these attributes do not depend on the entire primary key, which creates partial dependencies, violating 2NF. To fix this we split table 2 into multiple, smaller tables, table 3, table 4, table 5, and table 6, in which the primary key is not a composite key. This way partial dependencies cannot occur.

shelter_id	shelter_name	shelter_address	shelter_capacity_current	shelter_capacity_max
1	Happy Tails	MainStreet 1	60	100
2	City Paws	Citystreet 3	85	200

animal_id	animal_name	animal_species	animal_age	animal_sex	animal_breed	shelter_id
1	Max	dog	2 years	male	border collie	1
2	Gerard	cat	1 year	male	ragdoll	1
3	Luna	dog	7 years	female	golden retriever	2
4	Harold	bunny	9 years	male	californian	1
5	Simba	bunny	2 years	female	netherland dwarf	1
6	Kitty	cat	5 years	female	persian	1
7	Stella	dog	10 years	female	labradore	2

animal_id	intake_id	intake_date	shelter_id
1	3	03-03-2024	1
2	4	27-05-2025	1
3	5	05-03-2022	2
4	6	06-03-2026	1
5	7	06-10-2023	1
6	8	04-02-2026	1
7	9	11-01-2024	2

staff_id	staff_name	staff_role	shelter_id
1	Kiki	Vet	1
2	Lara	Vet	1
3	Zara	Vet	2
4	Chantal	Receptionist	1
5	Ariana	Assistant Vet	1
6	Harry	Caretaker	1
7	Benito	Animal Groome	2

After solving 2NF, 3NF is also solved already as there are no transitive dependencies.

Final ERD

